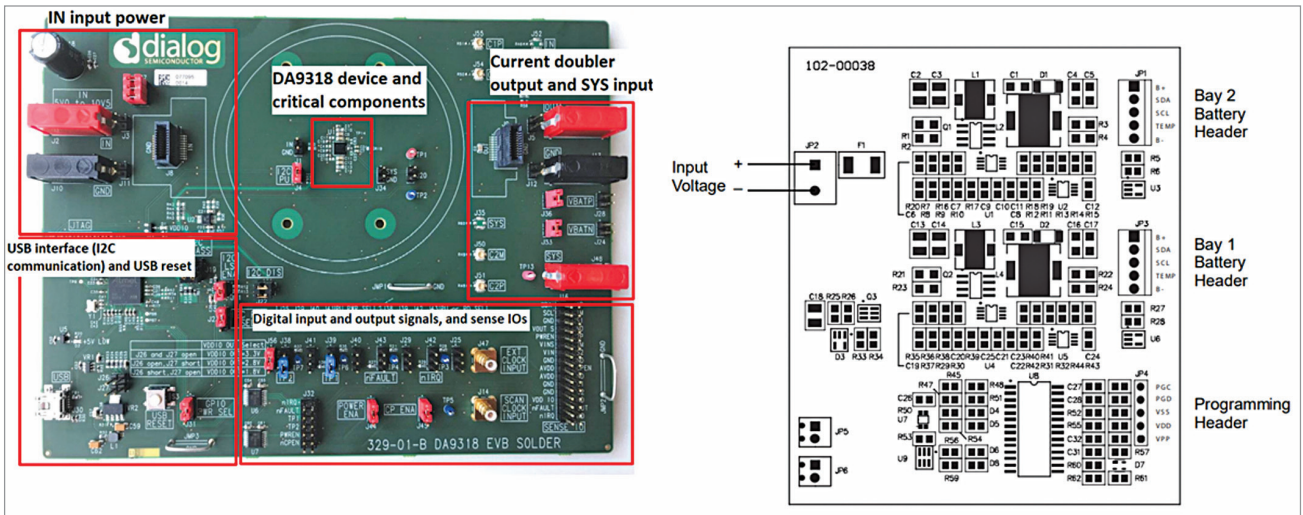


Li-ion Battery Charger REFERENCE DESIGNS

The main purpose of this article is to help design engineers in selecting the right battery management and charging IC and completing their proof of concept (PoC) phase. The resources provided include schematics, bill of material (BOM), design files, GUI interface software, etc



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The lithium-ion (Li-ion) cell is one of the most common energy sources in today's battery-driven world. Li-ion cells have good energy density and can be recharged. But they are very volatile and thus require specialised chargers to keep the battery and device safe.

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All the reference designs in the article can be used as independent cell chargers. Some of the designs are small enough to be integrated into a product as a sub-system for charging an in-built battery.

In this article, we will discuss the essential parameters of each of the Li-ion chargers. Table 1 lists the important parameters of all the charging reference designs.

TIDA-00590 for fast charging with low thermal budget

The TI TIDA-00590 is a fast charger reference design module from Texas Instruments. It is capable of fast charging a single Li-ion cell with 5A charge current with low thermal budget. The circuit design has been tested by TI and uses the EVM GUI and getting started guide.

What is a Reference Design?

Wikipedia defines a reference design as a technical blueprint of a system that is intended for others to copy. The main purpose of the reference design is to support companies in the development of next-generation products using the latest technologies. The reference product is proof of the platform concept and is usually targeted for specific applications. It enables a fast track to market, thereby cutting costs and reducing risk in the customer's integration project.